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## COMPARATIVE DEMOGRAPHY OF THREE SPECIES OF SCLERACTINIAN CORALS USING AGE- AND SIZE-DEPENDENT CLASSIFICATIONS<sup>1</sup>

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**Abstract.** Three species of massive reef-building coral, *Goniastrea aspera*, *G. favulus*, and *Platygyra sinensis*, were studied on two fringing reef flats in the central Great Barrier Reef from 1982 to 1984. Total population sizes ranged from 25 to 292 colonies, and remained relatively constant. Population structures and dynamics based on both age and size were described. Differences between these two classifications were primarily due to tissue fission or shrinkage of colonies. In populations dominated by small size classes, young corals were not necessarily the most abundant. Similarly, populations dominated by the largest size classes were not always dominated by the oldest corals. In several populations, mean colony size decreased slightly with increasing age, though variability in size also increased. Mortality patterns were similar in all three species, with the youngest genets and smallest ramets suffering the highest death rates. While mortality in older or larger corals was low, the incidence of partial mortality and fission was higher in these groups. Age-specific fecundity increased rapidly once reproductive age was reached at  $\approx 5$  yr, but fecundity decreased slightly in the older age classes due to the decrease in mean colony size. Estimated mean generation times ranged from 33 to 37 yr.

**Key words:** age-structured populations; coral demography; fecundity schedules; generation times; life histories; life tables; size-structured populations; survivorship.

### INTRODUCTION

The modular structure, indeterminate growth, and cloning of many plants and animals present serious difficulties for demographic analyses using approaches first developed for unitary organisms. These approaches are based on the ages of individual organisms (Deevey 1947, Keyfitz 1977). In modular organisms, however, the ages of genetic individuals (genets) may be difficult to determine, due to such factors as both growth and shrinkage, fission of genets into several parts (ramets), and fusion of ramets. Even the status of an organism (genet or ramet) can be uncertain. All these properties are possible because modular organisms have subunits that can function as physiologically separate entities. In corals, these units are polyps, which bud to form a colony. The polyps of the original colony may divide, at any stage and by a variety of processes, to form a clone consisting of a number of daughter colonies, each composed of some of the original polyps. Problems arising from these properties are particularly acute when rates of clonal propagation are high, however in many instances the relative importance of sexual vs. clonal propagation and the ages of organisms are unknown due to the difficulties in identifying ge-

netic individuals (Harper 1977, Hughes 1984, Hughes and Jackson 1985). As a consequence of these problems, size may be more relevant than age in understanding the demography of populations of clonal organisms such as corals. Recently, population ecologists working principally with plants have successfully addressed these problems using methods based on size rather than age (Werner and Caswell 1977, Hubbell and Werner 1979, Caswell 1982, Kirkpatrick 1984, Caswell 1985). Such methods are clearly relevant to modular marine invertebrates and have also been applied to corals (Hughes 1984). Despite the usefulness of these approaches, they overlook potentially important interactions between size and age (Werner and Caswell 1977, Chapman 1986, Hughes and Connell 1987), and the fact that populations of ramets and of genets may each have their own internal population dynamics, with quite different characteristics (Harper and Bell 1979).

While size-dependent models of populations are useful in understanding factors affecting dynamics and fitness in populations (Caswell 1982), these models alone do not incorporate information on the rates of genetic change and the evolution of populations (Hickman 1979). Such information is essential in any interpretation of an organism's evolutionary history, such as, for example, attempts to compare patterns of speciation in clonal and aclonal groups of marine organisms (Potts 1983). Demographic studies that combine

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both age- and size-dependent approaches add significantly to any understanding of the dynamics of these populations. Such a combined approach has been used in a variety of organisms (Werner 1975, Van Der Meijden and Van Der Waals-Kooi 1979, Gross 1981, Chapman 1986, Peterson 1986), including corals, in which both size and age can affect growth and mortality rates (Hughes and Connell 1987).

The ability of genets of modular organisms to propagate clonally into two or more physically separate parts has several important consequences. Cloning may decrease or increase the probability of death of the genotype, depending on whether the resulting ramets have higher or lower probabilities of surviving than the parental ramet (Cook 1979). Many plants and marine organisms possess adaptations for vegetative propagation, which serve to maximize the advantages of literally "spreading" the risk of mortality (Harper 1977, Highsmith 1982, Karlson 1986). In addition to direct effects on the chances of survival of a genotype, cloning or fission may also influence a genet's fitness by altering its reproductive output. Fission in clonal organisms that do not possess special adaptations often involves an initial net loss in size, with daughter colonies being formed by the death of intervening parts of the parent colony. In types of coral that do not possess such adaptations this loss is often major (Hughes and Jackson 1985). Since size can be an important determinant of reproductive output in organisms like most plants (Harper 1977) and corals (Kojis and Quinn 1985, Szmant-Froelich 1985), immediate reproductive output may be reduced significantly (albeit temporarily). Lifetime reproductive output may not be increased over that which might have been achieved by the original undamaged organism, unless the chances of survival for resulting ramets are extremely high.

The corals *Goniastrea aspera* Verrill, *G. favulus* (Dana), and *Platygyra sinensis* (Edwards and Haime) build "massive" skeletons (approximately hemispherical domes), which remain intact and contain evidence of the history of the coral. For this reason genets can be defined, and ages can be estimated, so that both age- and size-dependent population parameters can be obtained. The probability of failing to recognize ramets is minimal because the massive skeletons remain intact, grow slowly, and attain relatively small sizes. Therefore these species are suitable for exploring how the decoupling of size and age influences demographic properties, particularly fecundity schedules. These species share similar habitats and growth forms, as well as close taxonomic affinities (Family Faviidae), but differ in aspects of their reproductive biology such as spawning behavior (Kojis and Quinn 1981, Babcock 1984). Life history parameters (egg size, fecundity, age at first reproduction) were compared to assess whether these traits were associated with different demographic properties of the species such as survivorship, colony size, and mean generation time.

## METHODS

### Study sites

Populations of *Goniastrea aspera*, *G. favulus*, and *P. sinensis* were studied on the fringing reef flats of two inshore high islands on Australia's central Great Barrier Reef, Geoffrey Bay at Magnetic Island (19°10' S, 146°51' E) and Pioneer Bay at Orpheus Island (18°35' S, 146°29' E). Reef flats at both sites are regularly exposed for up to several hours at spring low tides (spring tidal range 3.5 m), and are situated in relatively turbid coastal waters. Detailed descriptions of the sites have been published by Morrissey (1980) and Hopley et al. (1983).

### Annual population surveys

Population counts and measurements were carried out between June and August in 1982, 1983, and 1984, during afternoon spring low tides. Sample populations were the total number of corals present in three, 1-m wide transects at each of the sites. The transects were 30 m long at Geoffrey Bay and 16 m long at Pioneer Bay, and were marked permanently by steel stakes driven into the reef flat at 5-m intervals. The higher density of corals at Pioneer Bay meant that shorter transects provided adequate numbers for that site. A tape measure was stretched along the stakes, and a 1-m<sup>2</sup> quadrat positioned astride the tape at 1-m intervals to form a continuous belt of quadrats, which could be re-established each year. Corals were recorded only if their centers fell within the quadrats. A number was assigned to each coral, and its species, quadrat number, and size were recorded. Thus, the fate (growth, fission, damage, mortality) of each coral (ramet and genet) could be followed over time. Two kinds of recruitment were recorded, larval recruitment and the formation of new ramets by tissue fission (death of intervening parts of larger colonies). Recruits originating from larval settlement were defined as small colonies ( $\approx 1$  cm diameter), not obviously originating from fission, which appeared in the quadrats in 1983 and 1984.

Measurements of colony size varied with the growth form of the coral. For hemispherical colonies (Fig. 1A) these were the height of the colony, the greatest basal diameter, and the diameter at right angles to that. For microatolls (Fig. 1B) or fragments of microatolls, two sets of measurements were made, since microatolls often have large areas of skeleton that are no longer covered by living tissue. The area of living coral was obtained from the outer circumference or length of the strip of living tissue around the perimeter, and the mean height of the living portion of the colony. In addition two diameters of the microatoll were measured (maximum diameter and that perpendicular to it). The arithmetic mean radius of the skeleton of every coral was calculated from diameter and height measurements. Surface areas were approximated either as the surface area of a hemisphere or as a rectangle for

microatolls with relatively two-dimensional tissue surfaces.

Data on the growth form of the coral was also recorded, such as whether the colony was an intact hemispherical form or a microatoll or some other damaged form. Where colonies were the product of tissue fission, this was recorded to establish both the number of ramets and the number of genets. This was possible since the skeleton usually remained unfragmented even though tissues may have separated due to death of intervening parts. Fusion between tissues of daughter colonies was recorded, as were cases of skeletal fusion between genetically distinct colonies in which the tissues remained separate (Collins 1978). Some fission events may have been obscured by subsequent fusion if both processes took place between annual censuses.

#### Growth rates

**Juveniles.**—Juvenile growth rates of *Goniastrea aspera*, *G. favulus*, and *Platygyra sinensis* were obtained using larvae produced during the annual spawning period of these species at each site in 1982, 1983, and 1984. Juveniles that settled on cut coral plates in the laboratory were returned to the reef and their growth was monitored. The methods used are detailed elsewhere (Babcock 1985). Juveniles were measured at intervals of several months, for a period of 14–15 mo after settlement. By this time most of the juveniles had died.

**Adults.**—Corals adjacent to the transects were stained in situ using alizarin dye techniques (Lamberts 1978) to establish whether growth rate varied with size and to confirm the annual periodicity of skeletal density banding. The corals stained were selected to cover as wide a size range as possible (1.5–17 cm mean radius). After 13–15 mo stained corals were collected and bleached to remove the tissue, then measured to determine mean radius. Bleached skeletons were sectioned vertically through the estimated origin of the colony, producing a central slab  $\approx 1$  cm thick. Growth after staining was measured at five equally spaced points around the perimeter of the colony, and the section was retained for x-ray examination.

Additional unstained corals of all three species were collected from both bays and were also sectioned. Sections from a total of 126 stained and unstained corals were then analyzed x-radiographically. Annual skeletal growth bands were measured from the negatives, starting from the most recent bands on the outside of the colony, and continuing towards the center of the colony until the bands became obscured, either by changes in growth orientation or by bioerosion. These provided annual growth increments for  $\approx 10$  yr prior to the study. Measurements were made with calipers from the center of each band along the axis with the longest continuous record of growth. Where banding was clear along more than one axis, up to three sets of measurements were made, and these were averaged for each year (Dodge

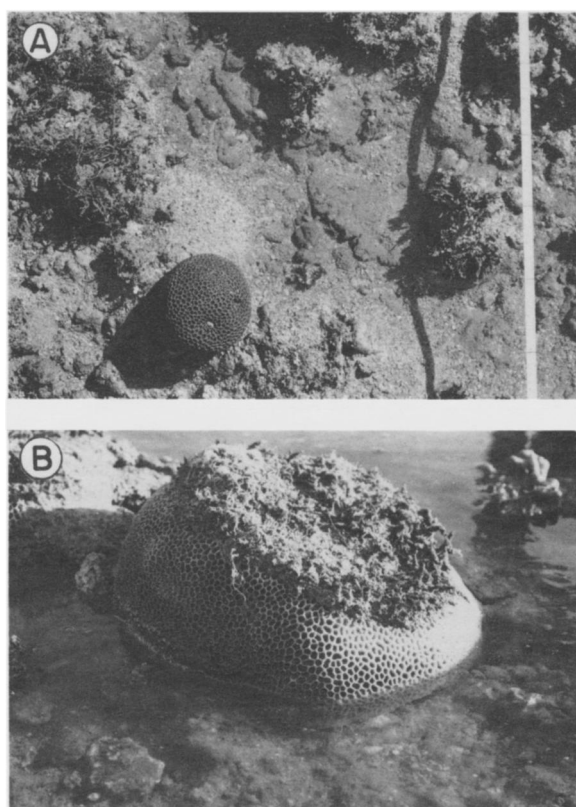


FIG. 1. (A) Typical hemispherical colony of *Goniastrea aspera* at Geoffrey Bay. (B) A microatoll of *Platygyra sinensis* showing dead upper surface of genet at Geoffrey Bay.

and Vaisnys 1977). Measurements were repeatable to  $\approx \pm 0.3$  mm (coefficient of variation = 0.03).

#### Age estimates

In conjunction with growth rate data, the skeletal measurements provided a means of ageing the corals. Ages of living corals were estimated using measurements of juvenile corals as well as growth rate data obtained from x-rays of adult corals. For the first 3 yr it was assumed that growth was largely two-dimensional and primary polyps continued to grow at the rate observed in juveniles over the 1st yr (until they reached the diameter of adult polyps,  $\approx 7$  mm). Once adult polyp size was reached growth was assumed to be three-dimensional and adult growth rates were applied linearly, since there was no evidence of size-dependent change in growth rate (see *Results: Growth rates and age estimates* [Table 4]). The age of each coral was estimated using the formula:  $\text{age} = 3 + [(r - 3j)/a]$ , where  $r$  = arithmetic mean radius of colony (in millimetres),  $j$  = mean annual radial growth rate of juveniles (first 3 yr), and  $a$  = mean annual adult growth rate.

#### Size-specific polyp fecundity

Fecundity in corals is a product of both number of polyps (area) and the fecundity of individuals polyps.

To establish how fecundity of single polyps changed with colony size, corals of a range of sizes, including both the largest and smallest colonies available, were sampled from each species adjacent to the study sites in 1983. The same data were also expressed as age-dependent polyp fecundity using ages estimated from skeletal dimensions. Corals chosen were all hemispherical in growth form in order to reduce the likelihood of possible effects on polyp fecundity in colonies that were remnants of older colonies or had suffered recent partial mortality. Samples ( $3 \times 3$  cm portions of tissue) were collected 3 wk before spawning at Geoffrey Bay, and 1 wk before spawning at Pioneer Bay. Approximately 50 colonies were sampled from each species at Geoffrey Bay, and 30 colonies from each species at Pioneer Bay. Sampling was deliberately biased towards small- to medium-sized corals to increase accuracy and precision in determining size at first reproduction.

Samples were fixed in 10% formalin and decalcified in 4% HCl (both volume/volume). Oocytes were counted in 15 mesenteries from each sample (5 mesenteries from each of 3 noncontiguous polyps), using a stereo dissecting microscope. Since the total reproductive output of a colony varies according to its surface area (the number of polyps increases as surface area increases), the mean number of polyps per square centimetre was measured as well as the average number of mesenteries per polyp. Values for polyp density were obtained from five bleached corals of each species, and the number of mesenteries per polyp counted in 10 polyps from each coral. These data were used to estimate average egg production per unit area for a colony of a given size, as well as in the construction of age-specific fecundity schedules. In calculating variances for estimates of egg numbers, all parameters were assumed to be independent, and the variance of the product of any two means calculated as  $(V_1/n_1)\bar{X}_2^2 + (V_2/n_2)\bar{X}_1^2 - (V_1/n_1)(V_2/n_2)$  (Travis 1982), where  $\bar{X}_1$  and  $\bar{X}_2$  are the mean values to be multiplied,  $V$  and  $n$  being their respective variances and number of values. This procedure was also used in calculating variances for other estimates which were the product of two means (such as survivorship and fecundity). The mean diameter of 20 freshly spawned eggs of each species was also measured to obtain comparative data on average diameters of mature eggs.

#### *General demographic assumptions*

In the case of the corals studied here, demographic calculations require that several assumptions be made: (1) 100% fertilization, (2) a well-mixed pool of larvae, (3) no density dependence, and (4) broadly representative populations. In the present case these assumptions are dealt with in the following ways. (1) The use of fecundity to estimate the zero age class is valid since all eggs of these corals are alive when they are spawned (Babcock 1986), and the probability of not being fer-

tilized is a legitimate source of mortality. (2) Most population models require that sexual reproduction results in new individuals entering the same local population. This would be an incorrect assumption to make in the case of corals such as those studied here, due to the dispersive nature of the eggs and larvae of these species (Babcock and Heyward 1986, Oliver and Willis 1987). Given the likelihood of larval dispersal and mixing on a scale comparable to that separating the study areas (Babcock 1988, Willis and Oliver 1988), populations from the two sites were combined to represent what is effectively a self-seeding "regional" population. While recruitment did vary between bays for the three species, this variability has been taken into consideration in the collection of the data. (3) The reef flats on which these corals are found are physically accommodated and densities of all species in this study are sufficiently low that density-dependent effects (intra- and interspecific interactions) are relatively insignificant. (4) While the populations are obviously in slightly different environments, there were no major catastrophic events such as cyclones, during the course of the study that might have affected one site differently from another. As for the sites being random samples this is of course not strictly true. Site selection was, however, haphazard, being essentially dependent on accessibility to humans. Finally, based on widespread personal experience in the region, the populations in this study are believed to be broadly representative of those found on the most common reef flat habitats of this region.

#### *Size class transition probability matrices*

Populations were divided into four arbitrary size classes, and the fate of corals in each size class was recorded annually over the period of study. The annual probability that corals would remain in the same size class after 1 yr, shrink or grow into other size classes, or die, were compiled into size-specific transition probability matrices (Werner and Caswell 1977, Hughes 1984). Sexual recruitment was incorporated into these matrices by using data obtained from each population regarding size-specific fecundity, total estimated annual reproductive output of the populations, and the annual recruitment rate. The mean fecundity per colony in each size class was calculated using the estimated size-specific fecundity of each coral. Fecundity values were incorporated in the matrix by creating a small nonreproductive size class, in which values represent the fecundities for the reproductive size classes. The probability of growing out of this size class (recruiting) or dying are calculated from the survival rate of larvae (calculated from the difference in total reproductive output and the mean annual recruitment rate for each species). The probability of staying in this size class cannot be given since corals of this size could not be observed *in situ*.

TABLE 1. Larval recruitment over 2 yr at Geoffrey and Pioneer Bays.

|                         | <i>Goniastrea aspera</i> |      |       | <i>Goniastrea favulus</i> |      |      | <i>Platygyra sinensis</i> |      |       |
|-------------------------|--------------------------|------|-------|---------------------------|------|------|---------------------------|------|-------|
|                         | 1983                     | 1984 | Mean  | 1983                      | 1984 | Mean | 1983                      | 1984 | Mean  |
| <b>Geoffrey Bay</b>     |                          |      |       |                           |      |      |                           |      |       |
| No. recruits            | 21                       | 24   | 22.5  | 8                         | 5    | 6.5  | 24                        | 12   | 18    |
| No. adults              | 180                      | 181  | 180.5 | 51                        | 49   | 50   | 102                       | 101  | 101.5 |
| Recruits/m <sup>2</sup> | 0.23                     | 0.26 | 0.25  | 0.09                      | 0.06 | 0.07 | 0.2                       | 0.13 | 0.2   |
| Adults/m <sup>2</sup>   | 2.0                      | 2.0  | 2.0   | 0.6                       | 0.5  | 0.6  | 1.1                       | 1.0  | 1.1   |
| <b>Pioneer Bay</b>      |                          |      |       |                           |      |      |                           |      |       |
| No. recruits            | 6                        | 10   | 8     | 35                        | 36   | 35.5 | 4                         | 1    | 2.5   |
| No. adults              | 50                       | 50   | 50    | 288                       | 288  | 288  | 29                        | 25   | 27    |
| Recruits/m <sup>2</sup> | 0.13                     | 0.22 | 0.17  | 0.8                       | 0.4  | 0.4  | 0.09                      | 0.02 | 0.03  |
| Adults/m <sup>2</sup>   | 1.1                      | 1.1  | 1.1   | 6.4                       | 6.4  | 6.4  | 0.6                       | 0.6  | 0.6   |

### Age-specific life tables

Age-specific mortality rates between 1982 and 1984 were used to calculate survivorship ( $l_x$ ) for each age class  $x$ . Corals in the 0–2 yr age class were too small to be observed directly. However, because both reproductive output and recruitment of populations were based on the same unit of area it was possible to estimate mortality from the difference in size of the zero age class (total fecundity of the populations at each site), and the mean annual recruitment rate (based on visible sexual recruits). The age-specific fecundity schedule for genets of each species was estimated directly from colony sizes and the size-specific fecundity for each population in 1984. Where genets consisted of a number of daughter colonies, the fecundity of each colony was calculated separately, and these were summed to provide the total fecundity of the genet. Age-specific fecundity schedules ( $m_x$ ) were then calculated as the mean fecundity of each age class for each population (Geoffrey and Pioneer Bays), and then pooled for both bays. For empty age classes,  $m_x$  values were estimated by averaging values of the four adjacent age classes, or of the two adjacent age classes if only one class was present on one side of the missing value.

## RESULTS

### Recruitment

Total sexual recruitment for *G. aspera*, *G. favulus*, and *P. sinensis* combined was similar at both sites, with a mean annual rate of 0.52 recruits/m<sup>2</sup> at Geoffrey Bay, only slightly lower than the 0.6 recruits/m<sup>2</sup> at Pioneer Bay. This was despite a roughly twofold difference in total adult density between the bays (Table 1). Recruitment and adult densities were positively correlated among the three species over the period of the study at both sites (Table 1; least squares regression,  $r = 0.894$ ,  $.05 > P > .01$ ,  $N = 12$ ).

### Growth rates and age estimates

In the first 14–15 mo following settlement juvenile corals of all three species achieved quite similar mean sizes (Table 2). Radial growth was slower in the first

6–8 mo than it was in the subsequent period, and the growth of juveniles was much slower than that of adult corals (Table 3). Among the adult corals, *P. sinensis* grew more rapidly than either of the two *Goniastrea* species, each of which grew at roughly similar rates (Table 3). Growth rate, as determined by alizarin staining, was independent of colony size in each of the species tested (Table 4). Comparison of alizarin staining bands and density banding patterns revealed by x-rays confirmed that pairs of one dense and one less dense band were laid down every year, justifying the use of density banding to measure annual growth.

Ages were estimated independently for each species at each site, using growth rates from x-ray data at that site. Fig. 2 presents graphically the relationships between size and age for each species and site. For the majority of the genets, i.e., those <20 yr old, standard errors are in the order of  $\pm 5$  yr; standard errors for the oldest members of each population are considerably larger (e.g.,  $\pm 10$  yr for a 40-yr-old coral).

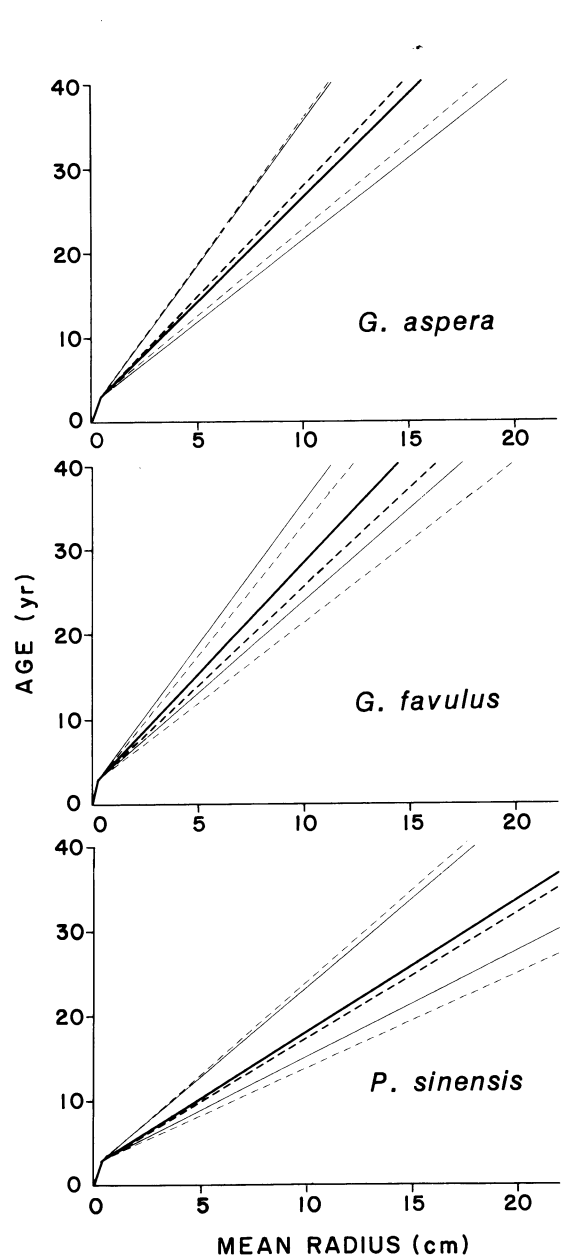
### Population structure

**Size distribution.**—There were consistent differences in population size structure and colony size between *Platygyra sinensis* and the two *Goniastrea* species at both bays (Fig. 3). All *Goniastrea* populations were dominated numerically by small colonies (<80 cm<sup>2</sup>), which made important but variable contributions to the total living surface area of the population, while *P. sinensis* populations were generally dominated by larger colonies, both numerically and in terms of their contribution to total cover. At each site the median colony size of *P. sinensis* was at least twice that of either *Goniastrea* species.

For each size class, the proportion of damaged colonies (i.e., microatolls, colonies resulting from tissue fission, or those having suffered substantial injury in the past) was highest in the largest size classes (Fig. 3). Damaged colonies were also present in most smaller size classes but at much lower frequencies. More *P. sinensis* colonies (82%) were damaged than in *G. aspera* or *G. favulus* (50 and 49%; Fig. 3). This may be because

TABLE 2. Diameter growth of juvenile *Goniastrea* and *Platygyra* corals (mm). *N* = number of juveniles. Diameter of newly settled corals  $\approx 0.45$  mm.

|                    | Age (mo)       |                |                |               |
|--------------------|----------------|----------------|----------------|---------------|
|                    | 3              | 5.5            | 7.8            | 14            |
| <i>G. aspera</i>   |                |                |                |               |
| Mean $\pm$ 1 SE    | 0.5 $\pm$ 0.02 | 0.9 $\pm$ 0.06 | 1.3 $\pm$ 0.09 | 2.8 $\pm$ 0.2 |
| ( <i>N</i> )       | (69)           | (98)           | (56)           | (3)           |
| <i>P. sinensis</i> |                |                |                |               |
| Mean $\pm$ 1 SE    | 0.5 $\pm$ 0.02 | 0.9 $\pm$ 0.05 | 1.2 $\pm$ 0.06 | 3.4 $\pm$ 0.8 |
| ( <i>N</i> )       | (79)           | (111)          | (72)           | (3)           |
| <i>G. favulus</i>  |                |                |                |               |
| Mean $\pm$ 1 SE    | 1.1 $\pm$ 0.2  | 1.5 $\pm$ 0.3  | 1.7 $\pm$ 0.3  | 3.2           |
| ( <i>N</i> )       | (16)           | (7)            | (4)            | (1)           |



*P. sinensis* colonies were larger and consequently more of these corals were likely to be microatolls, or to have suffered major damage at some time in the past. The proportion of ramets to genets in the population was also much greater in *P. sinensis* than in either *Goniastrea* species (Table 5). Pioneer Bay had higher proportions of small colonies than Geoffrey Bay, and all species also had smaller mean sizes there (Table 6). Overall there was also a higher proportion of damaged corals at Pioneer Bay (60%) than at Geoffrey Bay (45%), a trend especially noticeable in the smallest size class ( $<40$  cm<sup>2</sup>), where the proportion of damaged colonies was 41% in *G. favulus*, and 75% in *P. sinensis* at Pioneer Bay. Pioneer Bay has a topographically more complex structure than the reef flat at Geoffrey Bay.

*Age distribution.*—Shapes of age distributions (Fig. 4) often differed from those of size distributions for the same populations (Fig. 3), as did the distributions of the proportion of total living area contributed by each size class. *Goniastrea aspera* and *G. favulus* at Geoffrey Bay had numerically smaller contributions from the youngest age class than would be expected from the

TABLE 3. Annual growth rates of *Goniastrea* and *Platygyra* corals obtained from x-ray analysis of density banding.

|                    | No. colonies | No. bands | Mean (mm/yr) | Range (mm) | SD   |
|--------------------|--------------|-----------|--------------|------------|------|
| Geoffrey Bay       |              |           |              |            |      |
| <i>G. aspera</i>   | 26           | 302       | 4.1          | 2–9.5      | 1.14 |
| <i>G. favulus</i>  | 17           | 214       | 3.8          | 2–7.8      | 0.84 |
| <i>P. sinensis</i> | 19           | 209       | 6.4          | 3.1–10.9   | 1.61 |
| Pioneer Bay        |              |           |              |            |      |
| <i>G. aspera</i>   | 20           | 220       | 3.9          | 1.7–8.1    | 0.97 |
| <i>G. favulus</i>  | 25           | 153       | 4.3          | 2.2–8.6    | 1.05 |
| <i>P. sinensis</i> | 19           | 190       | 6.8          | 2.1–12.7   | 2.21 |

FIG. 2. Size–age relationships (means  $\pm$  1SE). Figure represents ages estimated by formulae using radial growth rates of skeleton for juveniles and adults (Tables 2 and 3). See *Methods: Age estimates* for method of estimation. Solid lines: Pioneer Bay; dashed lines: Geoffrey Bay.

TABLE 4. Colony size and growth rate. Linear regression of mean radius and mean radial growth rate for 13–15 mo, 1982–1983 (alizarin staining technique). *N* = number of colonies.

|                           | <i>N</i> | Size range (cm) | <i>R</i> <sup>2</sup> | <i>y</i> intercept | Slope | <i>P</i> |
|---------------------------|----------|-----------------|-----------------------|--------------------|-------|----------|
| <i>Goniastrea aspera</i>  | 39       | 1.5–12.6        | 0.02                  | 3.6                | −0.01 | >.1      |
| <i>G. favulus</i>         | 33       | 1.6–8.0         | 0.03                  | 4.5                | −0.01 | >.1      |
| <i>Platygyra sinensis</i> | 31       | 2.6–16.8        | 0.02                  | 5.7                | −0.01 | >.1      |

size-frequency distribution, and intermediate rather than young age classes were the most abundant. However at Pioneer Bay, the strong dominance of the smallest size classes was also reflected in dominance of the youngest age classes. For *Platygyra sinensis* at Geoffrey Bay, and to a lesser extent at Pioneer Bay, older age

classes were less abundant than would seem to have been indicated by the bimodal size-frequency distribution.

The majority of total area of live coral for all populations was concentrated in a few middle age classes (Fig. 4), in distinct contrast to the much broader distributions for size structures. For example, middle age classes contributed most of the tissue area of the *Platygyra* population at Geoffrey Bay, yet in terms of size structures, living area was totally dominated by corals in the largest size classes. There were similar differences between age and size structures among the younger age classes of *Goniastrea* populations at Pioneer Bay. There, the smallest size classes clearly made the greatest contribution to the proportion of total live area, yet this was not true for the age-dependent population structure, where the youngest genets made only a minor contribution to cover despite their numerical dominance (Figs. 3 and 4).

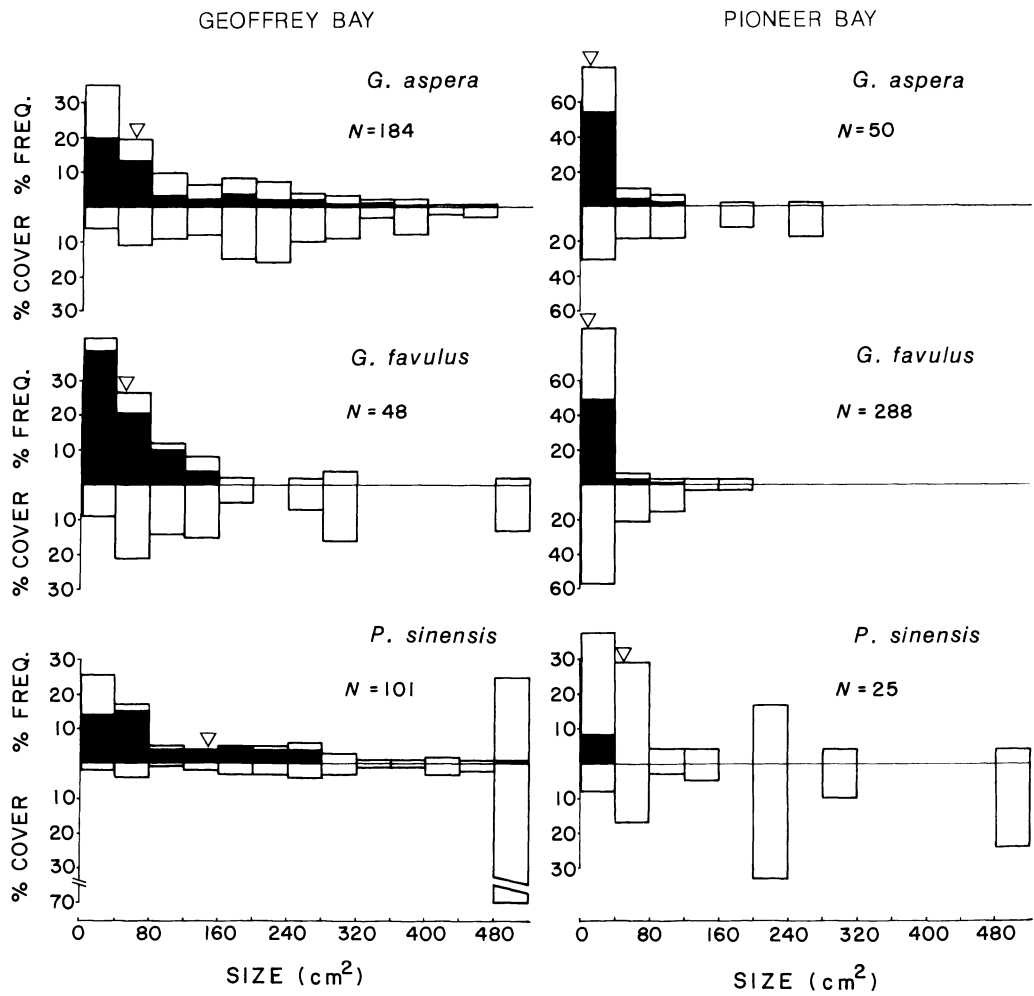


FIG. 3. Size-specific frequency distributions of ramets and total cover. Solid portions of the bars above horizontal axis are proportions of undamaged hemispherical colonies, open portions are frequencies of damaged colonies, microatolls, or colonies resulting from fission. Bars below horizontal axis are proportions of total cover (living surface area of the population) contributed by each size class in 1984. Arrows indicate median colony size. *N* = number of colonies.

TABLE 5. Population sizes, mortality, and recruitment of *Goniastrea* and *Platygyra* corals. Numbers of colonies as ramets and genets (ratio of ramets to genets in parentheses); mortality refers only to mortality of the entire ramet and not to partial mortality of a colony; recruits are those new colonies in the populations formed either by fission (ramets) or by sexual recruitment (genets).

|                           | 1982     |       | 1982-1983 |     |         | 1983     |       | 1983-1984 |     |         | 1984     |       |
|---------------------------|----------|-------|-----------|-----|---------|----------|-------|-----------|-----|---------|----------|-------|
|                           | Colonies |       | Recruits  |     |         | Colonies |       | Recruits  |     |         | Colonies |       |
|                           | Ramet    | Genet | Died      | Sex | Fission | Ramet    | Genet | Died      | Sex | Fission | Ramet    | Genet |
| Geoffrey Bay              |          |       |           |     |         |          |       |           |     |         |          |       |
| <i>Goniastrea aspera</i>  | 202      | 183   | 49        | 21  | 8       | 182      | 147   | 25        | 24  | 3       | 184      | 164   |
|                           | (1.1)    |       |           |     |         | (1.23)   |       |           |     |         | (1.12)   |       |
| <i>G. favulus</i>         | 54       | 51    | 12        | 8   | 1       | 51       | 51    | 8         | 5   | 0       | 49       | 48    |
|                           | (1.06)   |       |           |     |         | (1.0)    |       |           |     |         | (1.02)   |       |
| <i>Platygyra sinensis</i> | 90       | 62    | 16        | 24  | 4       | 102      | 74    | 20        | 12  | 7       | 101      | 76    |
|                           | (1.45)   |       |           |     |         | (1.38)   |       |           |     |         | (1.33)   |       |
| Pioneer Bay               |          |       |           |     |         |          |       |           |     |         |          |       |
| <i>G. aspera</i>          | 53       | 44    | 10        | 6   | 1       | 50       | 42    | 10        | 10  | 0       | 50       | 46    |
|                           | (1.20)   |       |           |     |         | (1.19)   |       |           |     |         | (1.09)   |       |
| <i>G. favulus</i>         | 292      | 251   | 45        | 35  | 6       | 288      | 247   | 47        | 36  | 11      | 288      | 242   |
|                           | (1.16)   |       |           |     |         | (1.17)   |       |           |     |         | (1.19)   |       |
| <i>P. sinensis</i>        | 25       | 15    | 3         | 4   | 3       | 29       | 14    | 6         | 1   | 1       | 25       | 13    |
|                           | (1.67)   |       |           |     |         | (2.07)   |       |           |     |         | (1.92)   |       |

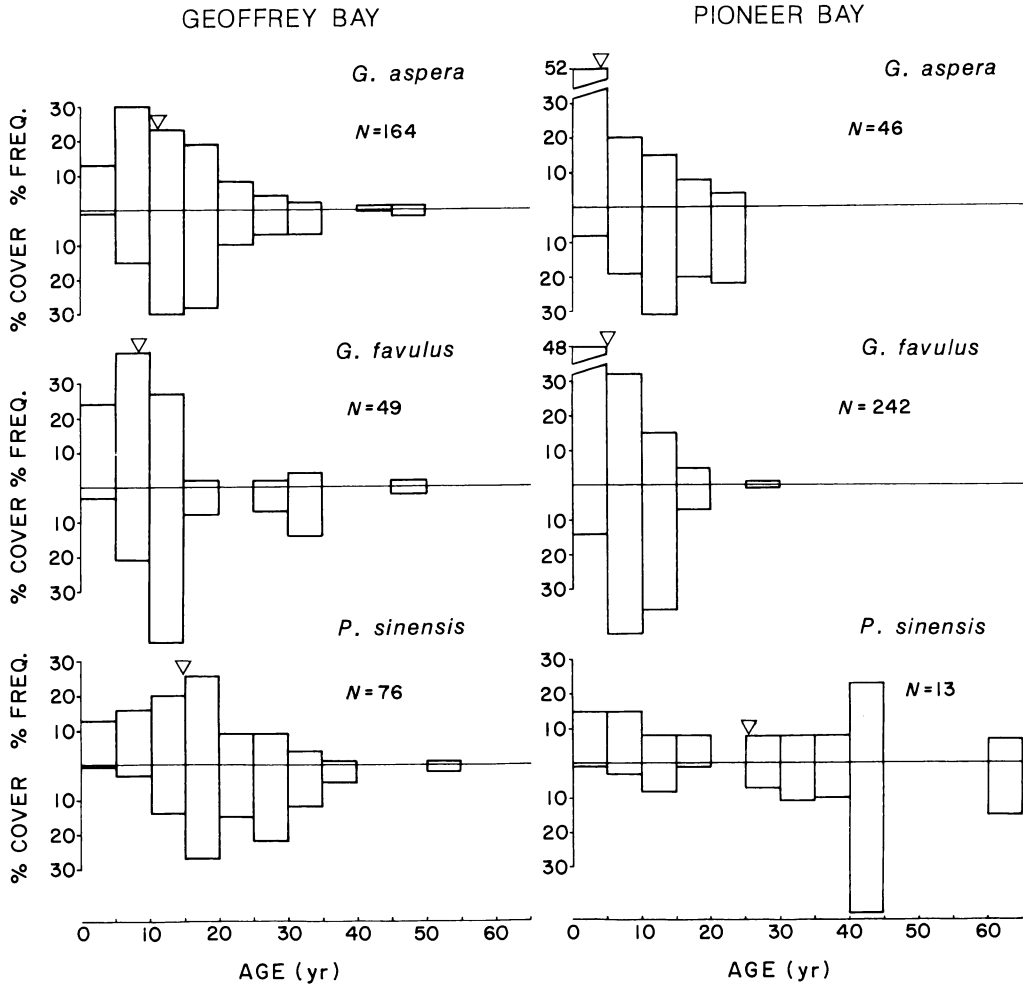


FIG. 4. Age-specific frequency distributions of genets and total cover. Histograms above horizontal axes are proportions of colonies in each age class: those below the axes are proportions of total living area contributed by each age class. *N* = number of genets. Arrows indicate median age.

Comparisons of size- and age-dependent population structures reveal several further differences. Numbers of genets were exceeded by numbers of ramets by between 9 and 48% (Table 5). In general terms, large corals were also older, however it was notable that mean colony size (living tissue of ramets) in the oldest age classes of several populations decreased, relative to mean colony size in the middle age classes (Fig. 5). It should also be noted that variances tended to increase with age, thus the values for mean size are the product of some very large, as well as some very small corals.

### Mortality

*Size-specific mortality.*—Patterns of mortality and turnover in the populations showed only relatively minor variations during the study, as might be expected in the absence of any major environmental disturbances such as cyclones. Whole-coral mortality rates for ramets were similar in all species, and at both bays during the period of the study. Overall mortality ranged

TABLE 6. Colony size and fission in *Goniastrea* and *Platygyra* corals. Formation of new colonies by tissue fission. Mean colony sizes are for 1984. The mean annual probability of fission was calculated by dividing the number of parent colonies observed by the total number of colonies in the population.

|                    | Mean colony size (cm <sup>2</sup> ) | Mean size parent colonies (cm <sup>2</sup> ) | Mean annual probability of fission |
|--------------------|-------------------------------------|----------------------------------------------|------------------------------------|
| Geoffrey Bay       |                                     |                                              |                                    |
| <i>G. aspera</i>   | 106                                 | 297                                          | 0.028                              |
| <i>G. favulus</i>  | 79                                  | 605                                          | 0.009                              |
| <i>P. sinensis</i> | 299                                 | 50                                           | 0.047                              |
| Pioneer Bay        |                                     |                                              |                                    |
| <i>G. aspera</i>   | 30                                  | 10                                           | 0.009                              |
| <i>G. favulus</i>  | 18                                  | 47                                           | 0.029                              |
| <i>P. sinensis</i> | 545                                 | 572                                          | 0.055                              |

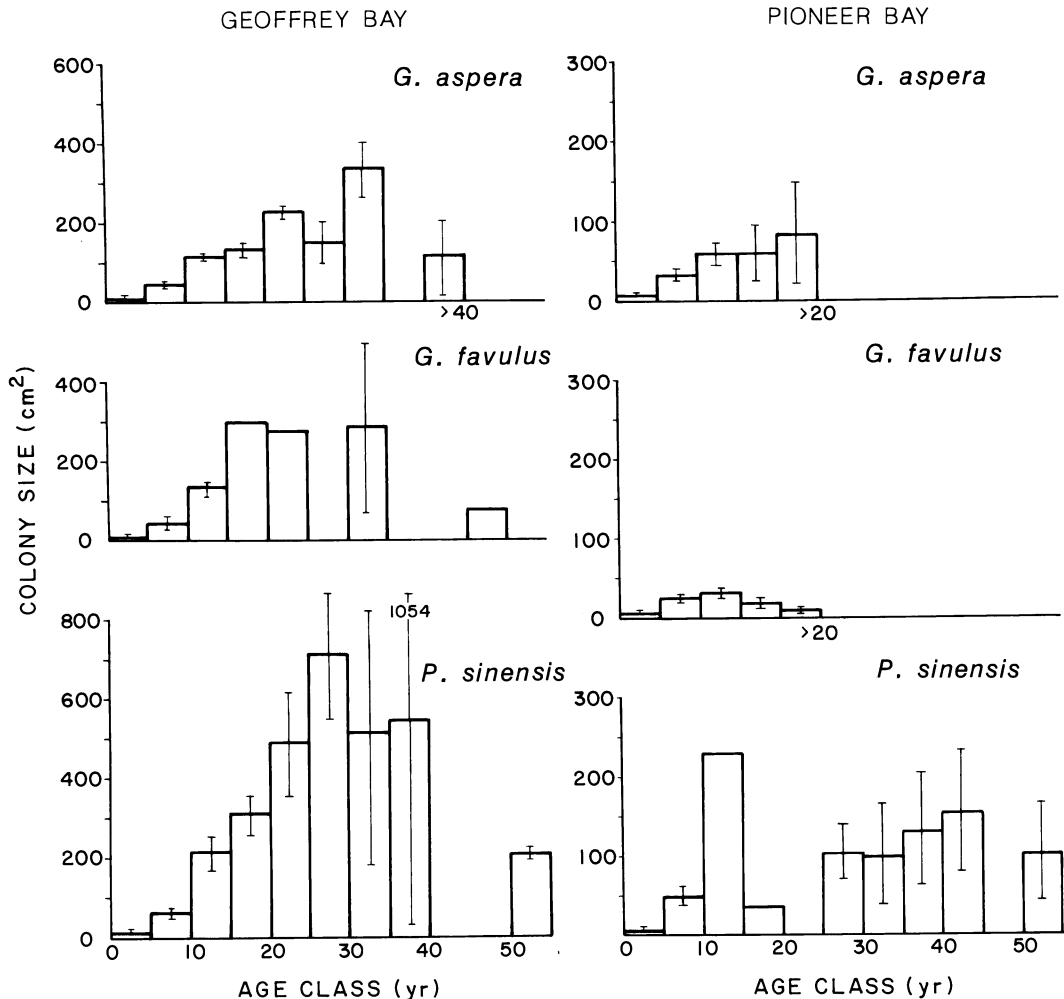


FIG. 5. Ramet size (area of living tissue) as a function of genet age (means  $\pm$  1 SE).

TABLE 7. Total live area and tissue turnover in *Goniastrea* and *Platygyra* corals. Area values are totals for tissue area of living coral in three transects at each bay.

|                    | 1982                         | 1982–1983     |               |                | 1983                         | 1983–1984     |               |                | 1984                         |
|--------------------|------------------------------|---------------|---------------|----------------|------------------------------|---------------|---------------|----------------|------------------------------|
|                    | Total area (m <sup>2</sup> ) | Area loss (%) | Area gain (%) | Net change (%) | Total area (m <sup>2</sup> ) | Area loss (%) | Area gain (%) | Net change (%) | Total area (m <sup>2</sup> ) |
| Geoffrey Bay       |                              |               |               |                |                              |               |               |                |                              |
| <i>G. aspera</i>   | 2.56                         | 40            | 17            | –23            | 1.95                         | 17            | 14            | –3             | 1.91                         |
| <i>G. favulus</i>  | 0.35                         | 25            | 25            | 0              | 0.35                         | 11            | 24            | +13            | 0.39                         |
| <i>P. sinensis</i> | 3.52                         | 26            | 13            | –13            | 3.09                         | 17            | 14            | –3             | 2.99                         |
| Pioneer Bay        |                              |               |               |                |                              |               |               |                |                              |
| <i>G. aspera</i>   | 0.14                         | 10            | 22            | +12            | 0.16                         | 19            | 13            | –6             | 0.15                         |
| <i>G. favulus</i>  | 0.66                         | 28            | 17            | –11            | 0.59                         | 30            | 16            | –14            | 0.50                         |
| <i>P. sinensis</i> | 0.43                         | 41            | 9             | –32            | 0.29                         | 18            | 21            | +4             | 0.28                         |

between 12 and 24% per year (Table 5), and no consistent trends were apparent between species or bays. Total ramet numbers in the sample populations decreased for the *Goniastrea* species at both Pioneer and Geoffrey Bays in 1982–1983, but remained virtually constant in 1983–1984. In contrast, ramet numbers of *P. sinensis* increased at both bays during 1982–1983, remaining constant at Geoffrey Bay in the following year, while dropping back to the 1982 level at Pioneer Bay. Proportionally, the highest annual mortality rates were recorded at Geoffrey Bay for *G. aspera* and *G. favulus*, and were caused by high mortality in the smallest size class in 1982–1983. The greatest increase in colony numbers was for *P. sinensis* during the same period and was caused principally by larval recruitment into the smallest size class rather than by fission (Table 5).

Changes in total live area (Table 7) for populations did not always correspond to changes in total colony number (Table 5). For example, although numbers of *P. sinensis* increased at both bays in 1982–1983, the total area of live coral decreased. Turnover in all populations was relatively high, with annual tissue loss due to partial or whole colony mortality ranging from 10 to 41%. The two greatest proportional losses of total cover both occurred in 1982–1983, in the population

of *P. sinensis* at Pioneer Bay (41%), and *G. aspera* at Geoffrey Bay (40%). The majority of losses in that year were caused by partial mortality in the largest size classes, with this contributing almost all of the loss to the *G. aspera* population, and just one colony contributing almost half the total loss to *P. sinensis*. The majority of populations showed a net tissue loss over the 2-yr period, however there were also cases of net increase in cover, due to a combination of growth, recruitment, and in rare cases, the immigration of unattached coral heads dislodged by storms or human interference.

Both colony size and previous damage to colonies had a significant influence on the probability of further damage in some populations (Table 8). The fate of colonies of each species was scored as either positive (growth) or negative (partial and whole colony mortality) and pooled for both years of the study. The frequency of positive and negative scores in each size class was then compared between previously undamaged colonies (hemispherical), and colonies that were noted to have sustained significant damage at some time in the past (microatolls, parts of microatolls, or other previously damaged colonies). For both *G. aspera* and *G. favulus*, previously damaged colonies in all size classes were more likely to sustain a loss in colony size than were undamaged hemispherical colonies (Table

TABLE 8. Influence of colony size and previous injury on growth or shrinkage of corals. Three-dimensional contingency tables examining the effects of size class and previous partial mortality on the probability of growth or shrinkage (including death). Size classes: I, visible—40 cm<sup>2</sup>; II, >40–80 cm<sup>2</sup>; III, >80–120 cm<sup>2</sup>; IV, >120 cm<sup>2</sup>. Data pooled for years and sites.

| Size class                                                | <i>Goniastrea aspera</i>              |        |                    |        | <i>Goniastrea favulus</i>             |        |                    |        | <i>Platygyra sinensis</i>             |        |                    |        |
|-----------------------------------------------------------|---------------------------------------|--------|--------------------|--------|---------------------------------------|--------|--------------------|--------|---------------------------------------|--------|--------------------|--------|
|                                                           | Uninjured                             |        | Previously injured |        | Uninjured                             |        | Previously injured |        | Uninjured                             |        | Previously injured |        |
|                                                           | Grew                                  | Shrank | Grew               | Shrank | Grew                                  | Shrank | Grew               | Shrank | Grew                                  | Shrank | Grew               | Shrank |
| I                                                         | 92                                    | 68     | 28                 | 48     | 214                                   | 148    | 78                 | 116    | 8                                     | 9      | 17                 | 30     |
| II                                                        | 28                                    | 12     | 8                  | 18     | 24                                    | 24     | 11                 | 12     | 3                                     | 2      | 8                  | 19     |
| III                                                       | 10                                    | 10     | 11                 | 14     | 13                                    | 14     | 1                  | 3      | 2                                     | 1      | 8                  | 13     |
| IV                                                        | 31                                    | 31     | 28                 | 45     | 4                                     | 6      | 2                  | 9      | 10                                    | 7      | 42                 | 54     |
| Mutual independence                                       | $\chi^2 = 45.1$ , df = 10, $P < .001$ |        |                    |        | $\chi^2 = 40.9$ , df = 10, $P < .001$ |        |                    |        | $\chi^2 = 40.9$ , df = 10, $P < .001$ |        |                    |        |
| Partial independence (growth on size and previous injury) | $\chi^2 = 23.3$ , df = 11, $P < .025$ |        |                    |        | $\chi^2 = 26.7$ , df = 11, $P < .01$  |        |                    |        | $\chi^2 = 6.0$ , df = 11, $P > .75$   |        |                    |        |

TABLE 9. Size-specific transition probability matrices. Data used to compile the matrices were combined for 1982–1983 and 1983–1984, and for populations at both bays. Size classes: I prior to visible recruitment; II visible–40 cm<sup>2</sup>; III >40–80 cm<sup>2</sup>; IV >80–120 cm<sup>2</sup>; V >120 cm<sup>2</sup>. *d* probability of dying.

| <i>Goniastrea aspera</i>  |                      |                   |                   |                   |                   |
|---------------------------|----------------------|-------------------|-------------------|-------------------|-------------------|
|                           | I                    | II                | III               | IV                | V                 |
| I                         | ...                  | $1.9 \times 10^3$ | $2.7 \times 10^4$ | $6.4 \times 10^4$ | $2.5 \times 10^5$ |
| II                        | $1.9 \times 10^{-6}$ | 0.58              | 0.21              | 0.04              | 0.05              |
| III                       | 0                    | 0.10              | 0.53              | 0.11              | 0.04              |
| IV                        | 0                    | 0                 | 0.12              | 0.49              | 0.12              |
| V                         | 0                    | 0                 | 0.02              | 0.27              | 0.75              |
| <i>d</i>                  | 0.9999981            | 0.32              | 0.11              | 0.11              | 0.05              |
| <i>Goniastrea favulus</i> |                      |                   |                   |                   |                   |
|                           | I                    | II                | III               | IV                | V                 |
| I                         | ...                  | $5.6 \times 10^2$ | $1.4 \times 10^4$ | $3.4 \times 10^4$ | $7.3 \times 10^5$ |
| II                        | $2.2 \times 10^{-5}$ | 0.77              | 0.28              | 0.13              | 0.1               |
| III                       | 0                    | 0.04              | 0.49              | 0.16              | 0.1               |
| IV                        | 0                    | 0                 | 0.13              | 0.46              | 0.3               |
| V                         | 0                    | 0                 | 0                 | 0.16              | 0.5               |
| <i>d</i>                  | 0.999978             | 0.18              | 0.1               | 0.09              | 0                 |
| <i>Platygyra sinensis</i> |                      |                   |                   |                   |                   |
|                           | I                    | II                | III               | IV                | V                 |
| I                         | ...                  | $4.8 \times 10^1$ | $4.5 \times 10^3$ | $5.7 \times 10^3$ | $1.8 \times 10^5$ |
| II                        | $1.9 \times 10^{-6}$ | 0.34              | 0.19              | 0.09              | 0.03              |
| III                       | 0                    | 0.17              | 0.48              | 0.46              | 0.03              |
| IV                        | 0                    | 0.02              | 0.06              | 0.27              | 0                 |
| V                         | 0                    | 0                 | 0.04              | 0.14              | 0.9               |
| <i>d</i>                  | 0.9999981            | 0.48              | 0.23              | 0.05              | 0.04              |

8). Undamaged colonies were more likely to grow, particularly in the smallest size classes, although this pattern was no longer visible or was even reversed in the larger size classes. The same trends were visible in *P. sinensis* although their significance could not be demonstrated, perhaps partially as a consequence of relatively low sample numbers.

The highest probability of whole colony mortality was always found in the smallest size class, ranging from 0.47 for *P. sinensis* to 0.18 for *G. favulus* (Table 9). The largest size classes tended to either remain in the same size class, or shrink down to one of the two smallest size classes. Fission of these large colonies frequently produced very small daughter colonies, while the major part of the original colony remained in the largest size class, e.g., *P. sinensis* (Table 9).

**Age-specific mortality.**—The recorded frequencies of genet mortality for corals in each age class were combined to estimate age-specific mortality rates for each population. Mortality was highest in the first age class, ranging from 21 to 69%, but dropped rapidly after 10 yr to relatively low values (Fig. 6). Variable mortality in the older age classes was likely to be the result of low sample sizes, particularly in the smaller populations. Net change in numbers of genets was relatively minor over the 2-yr period of the study, with both increases and decreases being recorded (Table 5). These changes largely parallel size-dependent mortality rates described above (*Results: Mortality: Size-specific mortality*) for ramets.

**Fission.**—The contribution of fission to “recruitment” was relatively minor in both *Goniastrea* species, but played a greater role in *Platygyra sinensis*, especially at Pioneer Bay (Table 6). For all populations the mean size of dividing colonies was much larger than the overall mean colony size, and large colonies appear to be more likely to divide than smaller ones. The mean annual probability of fission showed a direct correspondence with mean colony size for each pop-

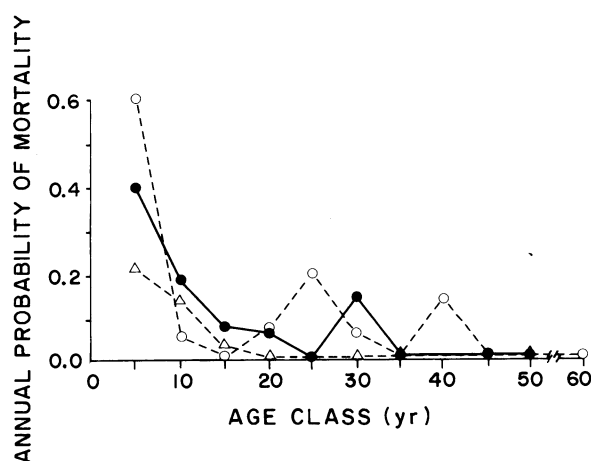


FIG. 6. Mean annual age-specific mortality of genets of ● *Goniastrea aspera*, ○ *G. favulus*, and △ *Platygyra sinensis*. Data were combined for the periods 1982–1983 and 1983–1984.

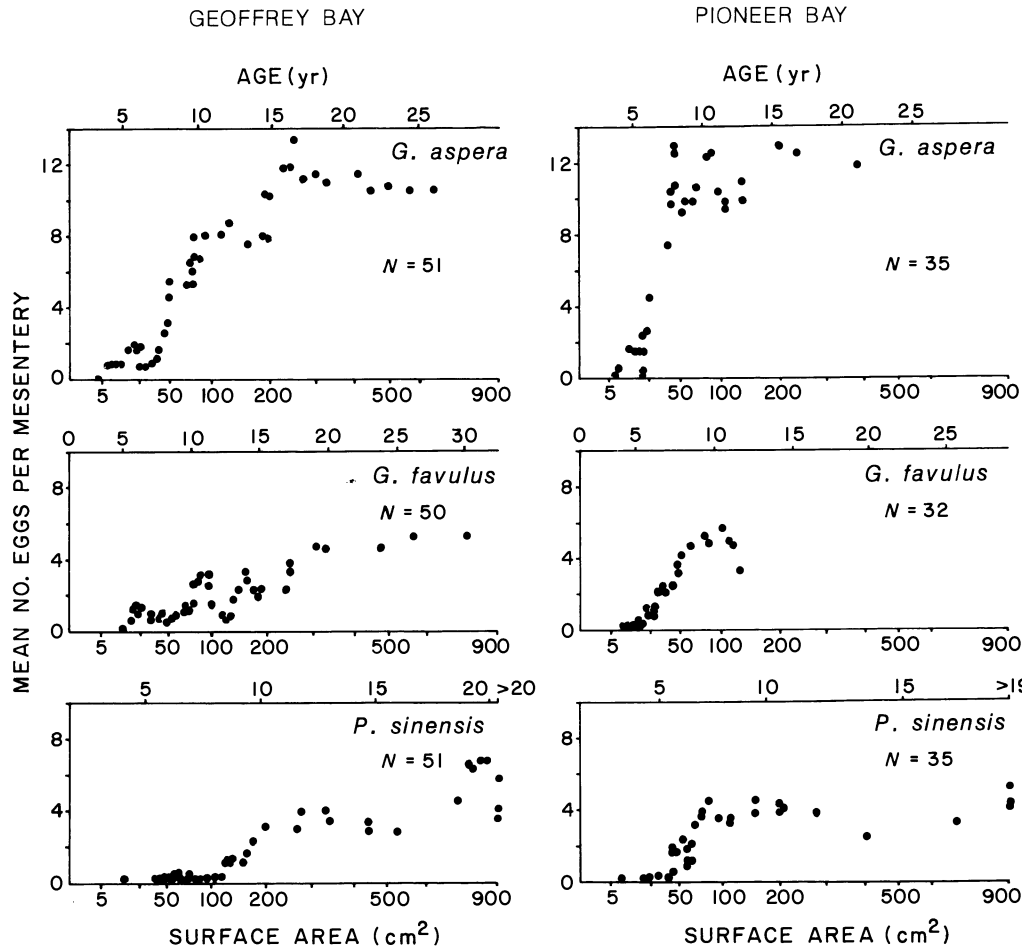


FIG. 7. Mean number of eggs per mesentery in relation to colony size (surface area) and estimated age. Plot of five-point running means. Surface areas are plotted as square root transformations of the actual values. *N* = number of colonies.

ulation (Table 6,  $r^2 = 0.73$ ,  $P < .01$ ), except *G. favulus* at Pioneer Bay, which had a relatively high probability of colony fission considering the very small mean colony size. The ratio of ramets:genets in populations of the three species (Table 5) closely reflected the rates of fission observed during the study ( $r^2 = 0.72$ ,  $P < .01$ ). *Platygyra*, which had the largest mean colony size, con-

sistently had the highest ratio of ramets to genets, as well as the highest recorded fission rates.

Fusion of daughter colonies was a rare occurrence, and only three instances were observed, one in each species. All of these fusions were between previously isolated parts of microatolls. Skeletal fusion (as opposed to tissue fusion) of unrelated corals of the same

TABLE 10. Fecundity of mature *Goniastrea* and *Platygyra* coral colonies. Fully mature colonies were assumed  $>200$  cm<sup>2</sup>, except *G. favulus* at Pioneer Bay ( $>100$  cm<sup>2</sup>). *N* = number of colonies from which estimates of no. eggs per mesentery were obtained. Data are means  $\pm$  1 SE.

|                    | <i>N</i> | No. of eggs per mesentery | No. septa per polyp | Polyps/cm <sup>2</sup> | Eggs/cm <sup>2</sup> |
|--------------------|----------|---------------------------|---------------------|------------------------|----------------------|
| Geoffrey Bay       |          |                           |                     |                        |                      |
| <i>G. aspera</i>   | 8        | 11.7 $\pm$ 1.1            | 23.7 $\pm$ 0.8      | 3.4 $\pm$ 0.9          | 940 $\pm$ 260        |
| <i>G. favulus</i>  | 11       | 3.9 $\pm$ 0.6             | 23.2 $\pm$ 1.0      | 3.9 $\pm$ 1.2          | 340 $\pm$ 130        |
| <i>P. sinensis</i> | 17       | 4.0 $\pm$ 0.8             | 20.0 $\pm$ 0.7      | 4.1 $\pm$ 0.7          | 330 $\pm$ 100        |
| Pioneer Bay        |          |                           |                     |                        |                      |
| <i>G. aspera</i>   | 4        | 11.2 $\pm$ 1.0            | 23.7 $\pm$ 0.8      | 3.4 $\pm$ 0.9          | 900 $\pm$ 250        |
| <i>G. favulus</i>  | 7        | 4.9 $\pm$ 0.8             | 23.2 $\pm$ 1.0      | 3.9 $\pm$ 1.2          | 420 $\pm$ 160        |
| <i>P. sinensis</i> | 8        | 3.6 $\pm$ 0.8             | 20.0 $\pm$ 0.7      | 4.1 $\pm$ 0.7          | 290 $\pm$ 100        |

species was observed to have occurred commonly in the past, although no new instances were observed. Since the tissues of these individuals do not fuse, the corals were also functionally separate and were always scored as separate genets.

#### Age at first reproduction

*Platygyra sinensis* first became reproductive at larger sizes ( $\approx 50 \text{ cm}^2$ , or 2.8 cm mean radius) than did either *Goniastrea* species (15–25  $\text{cm}^2$ , or 1.5–2 cm mean radius), but since *P. sinensis* grows more rapidly (Table 3), the age at which each species began reproductive activity was similar (Fig. 7). This age did vary, however, commencing at 4–5 yr in the most precocious species *G. aspera*, and ranging from 5 to 8 yr in *P. sinensis*.

#### Fecundity

The numbers of eggs per mesentery reached by mature *G. aspera* (11–12) were approximately twice those found in *G. favulus* (4–5) and *P. sinensis* (3–4) (Fig. 7). In general, adult levels of polyp fecundity (Table 10) were reached by each species at sizes of between 100 and 200  $\text{cm}^2$ . This was equivalent to an age of between 10 and 15 yr, even though there was some variation between bays. For example, in each species maximum numbers of eggs per mesentery were reached at smaller colony sizes and younger ages at Pioneer Bay than at Geoffrey Bay, although maximum numbers were similar at both bays (Fig. 7).

#### Size class transition matrices

Size-specific transition probabilities are summarized in Table 9. The top row of each matrix contains annual size-specific ramet fecundities. The main diagonal (upper left to lower right) contains probabilities of remaining the same size; entries above and below this diagonal are probabilities of shrinking to smaller sizes or growing to larger sizes, respectively. The most likely fate of colonies in all size classes was to remain in the same size class, while the next most likely outcome was to move into one of the two adjacent (larger or smaller) size classes. Fecundity of colonies was highest in the largest size class for each species, with the three smallest size classes having very low fecundities.

#### Age-specific life tables

**Age-specific survivorship.**—Survivorship curves for genets (Fig. 8, Tables 11–13) were of the same general form in *G. aspera*, *G. favulus*, and *P. sinensis*, with the large differential between annual reproductive output (0 age class) and annual recruitment to visible size indicating a high rate of larval and juvenile mortality. Levels of mortality in *G. aspera* and *P. sinensis* were slightly higher overall than that of *G. favulus*. This difference appears to be due largely to a relatively lower level of mortality in *G. favulus* in the first age class, prior to visible recruitment. Differences in mortality

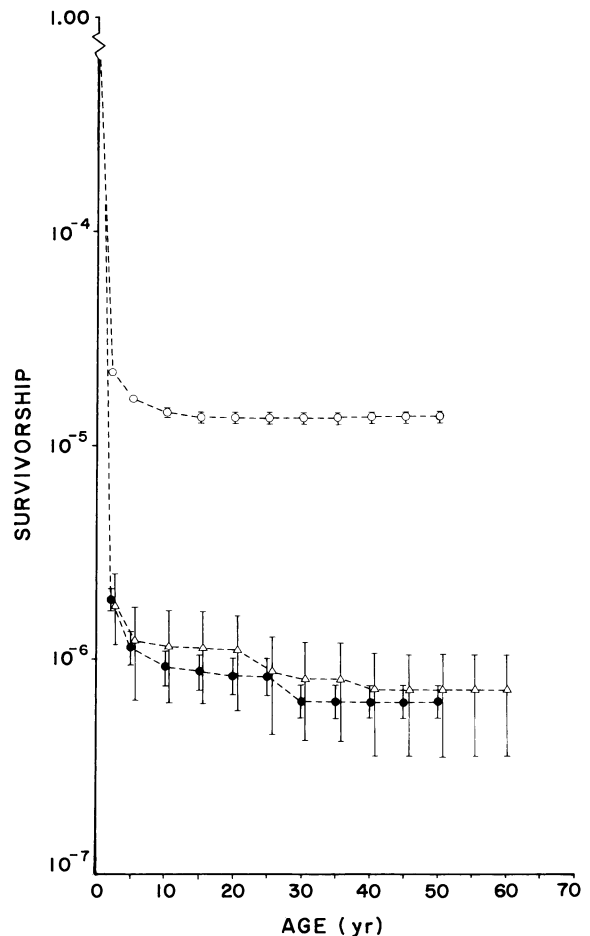


FIG. 8. Age-specific survivorship of genets of ● *Goniastrea aspera*, ○ *G. favulus*, and △ *Platygyra sinensis* (pooled populations): means  $\pm$  1 SE.

could occur at several stages during the presettlement period, i.e., at fertilization, prior to settlement, or after settlement but prior to reaching visible size. Once corals reached visible size, survivorship was much higher than in the earlier stages, but mortality was nevertheless greater in the youngest genets, tapering off to low levels for the oldest members of the populations. Survivorship in the oldest age classes as observed over the period of the study (Fig. 6) indicated very low rates of mortality of these genets. However, it should be noted that the oldest corals in the various populations were not immune to the processes of mortality and were subject to factors such as partial mortality and fission (Tables 6 and 8).

**Fecundity.**—Age-specific fecundity schedules for *Goniastrea aspera*, *G. favulus*, and *P. sinensis* showed rapid increases in total fecundity after reproductive age was reached, due to a combination of increased polyp fecundity and increased surface area. The general forms of the  $m_x$  curves presented here are basically similar (Fig. 9), however, that of *G. favulus* rises less steeply than that of the other two species. This is a result of

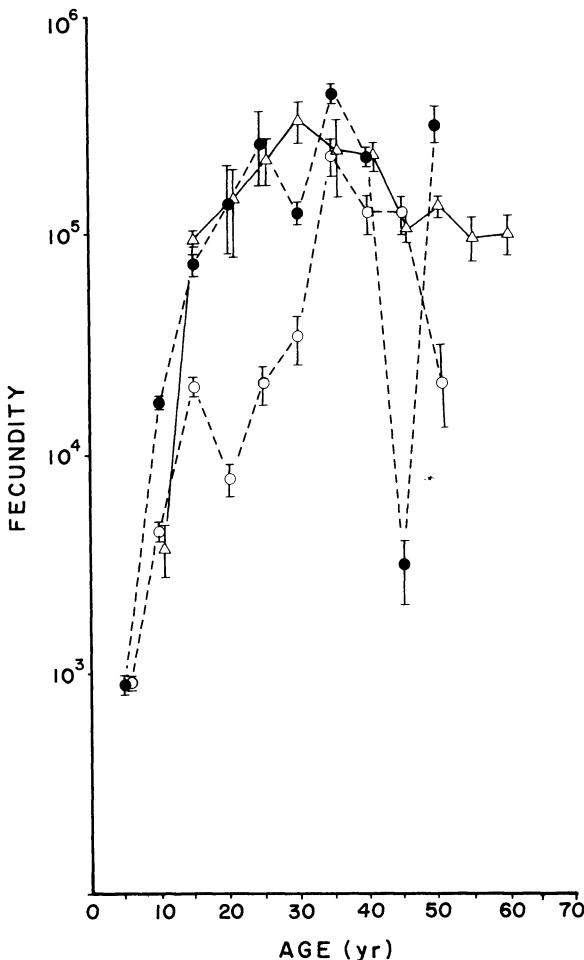


FIG. 9. Age-specific fecundity schedules of ramets of ● *Goniastrea aspera*, ○ *G. favulus*, and △ *Platygyra sinensis* (pooled populations): means ± 1 SE.

both low polyp fecundity and small size. *Platygyra sinensis*, a larger coral with a faster growth rate and similar polyp fecundities, increased total reproductive output at about the same rate as *G. aspera*, after commencing reproduction at a slightly older estimated age. The average reproductive output per genet tended to be greatest in the middle age classes, with some of the oldest corals often having significantly reduced fecundity. Fecundity decreased in older genets as a result of decreases in mean surface area of these corals or the ramets of which they were composed. The oldest age classes did contain some genets with very high fecundities, however they also contained the highest proportions of colonies that were the product of tissue fission and partial mortality. Mean generation times were similar in the three species, ranging from 33 yr in *G. aspera* to 37 yr in *P. sinensis* (Tables 11–13).

Age- and size-dependent classifications (based on genets and ramets, respectively) performed similarly in terms of the resulting population structures and patterns in mortality and fecundity. In some cases differ-

ences between the two approaches were simply inherent in the different methods (such as the inability of measures based on size alone to provide a measure of generation times, or the inadequacy of age-specific life tables in dealing with fission). Used independently, both measures provided adequate general descriptions of the populations in question, but the use of age- and size-dependent measures in combination added significantly to the understanding of the populations. Comparisons of genet age and ramet size, for example, revealed a mean decrease in size with increasing age in several of the populations. This trend was echoed in the data for age-specific fecundities, since size was a factor implicit in calculating the fecundity values.

## DISCUSSION

### *Population structure and dynamics*

A number of differences were apparent between population structures based on size- and age-dependent measures. The size-structured populations of *P. sinensis*, especially at Geoffrey Bay, were dominated by corals in the largest size classes, both numerically and in terms of their contribution to the total surface area covered by the population, yet in the age-structured populations it was corals in the middle age classes, not the oldest corals, which dominated. In an example from the younger portion of the populations, the smallest size class was most abundant in both *Goniastrea* species at Geoffrey Bay, however in the age-structured populations the youngest age class was less numerous than several of the older age classes (Figs. 3 and 4). Thus, even in these species with massive growth forms, size structure of the population was significantly affected by shrinkage resulting from partial mortality or tissue fission. In all populations virtually every size class contained at least some colonies that had been larger at some previous point in time (Fig. 3). This was especially true in populations of *Platygyra sinensis*, which attains a larger size than either *Goniastrea* species, increasing its chances of partial mortality and fission (Tables 6 and 10). Furthermore, once injured, the likelihood of further injury or death was greater than for colonies that had not been injured in the past, a trend which cannot be solely attributed to differences in size (Table 8).

For reef-flat species such as *G. aspera*, *G. favulus*, and *P. sinensis* the probability of some kind of damage occurring to larger and usually older corals increased for a number of reasons. Larger colonies project farther above the reef flat and are exposed at low tides more frequently and for longer periods of time. They are thus more likely to become microatolls. Random deleterious events are also more likely to affect colonies as they grow and cover more space. Bioerosion of the substratum or the skeleton of the coral at the attachment point may also increase over time resulting in the dislodgment of colonies. These environmental con-

TABLE 11. Life table for the coral *Goniastrea aspera*. Size of the 0 age class is based on the estimated fecundity of the populations.  $f_x$  = frequency of genets in each age class; — = too small to be visible in a field census; \* = values estimated from adjacent age classes. See Figs. 8 and 9 for standard error of  $l_x$  and  $m_x$ .

| $x$<br>(age class)                                                                                              | $f_x$              | $l_x$                 | $m_x$    | $l_x m_x$<br>( $\bar{X} \pm 1$ SE) |
|-----------------------------------------------------------------------------------------------------------------|--------------------|-----------------------|----------|------------------------------------|
| 0                                                                                                               | $1.57 \times 10^7$ | 1.00                  | 0        | 0                                  |
| 2                                                                                                               | —                  | $1.94 \times 10^{-6}$ | 0        | 0                                  |
| 5                                                                                                               | 45                 | $1.1 \times 10^{-6}$  | 909      | .00099 $\pm$ .0002                 |
| 10                                                                                                              | 52                 | $0.93 \times 10^{-6}$ | 17 900   | .0166 $\pm$ .0033                  |
| 15                                                                                                              | 45                 | $0.89 \times 10^{-6}$ | 74 200   | .066 $\pm$ .015                    |
| 20                                                                                                              | 34                 | $0.85 \times 10^{-6}$ | 142 000  | .121 $\pm$ .054                    |
| 25                                                                                                              | 16                 | $0.85 \times 10^{-6}$ | 276 000  | .235 $\pm$ .054                    |
| 30                                                                                                              | 6                  | $0.64 \times 10^{-6}$ | 125 000  | .080 $\pm$ .032                    |
| 35                                                                                                              | 2                  | $0.64 \times 10^{-6}$ | 456 000  | .292 $\pm$ .084                    |
| 40                                                                                                              | 0                  | $0.64 \times 10^{-6}$ | 229 000* | .146 $\pm$ .031                    |
| 45                                                                                                              | 1                  | $0.64 \times 10^{-6}$ | 3140     | .002 $\pm$ .0007                   |
| 50                                                                                                              | 1                  | $0.64 \times 10^{-6}$ | 330 000  | .212 $\pm$ .056                    |
| $\Sigma l_x m_x = 1.17 \pm 0.014$                                                                               |                    |                       |          |                                    |
| $r = 0.005$                                                                                                     |                    |                       |          |                                    |
| generation time ( $G$ ) = 33 + 11 yr, where                                                                     |                    |                       |          |                                    |
| $G = \Sigma x l_x m_x / \Sigma l_x m_x$ ; $r$ = intrinsic rate of increase estimated by iteration (Krebs 1972). |                    |                       |          |                                    |

straints on continued growth may be equivalent in their demographic consequences to a period of "senescence" in these species. A comparison of size and age structure for each population has revealed that as a result of these factors, the older members of some populations did not continue to grow in mean size but appeared to suffer a mean trend towards a reduction in surface area of living tissue (Fig. 5). On theoretical grounds, senescence is not expected to occur in colonial organisms such as clonal plants and corals (Harper 1980). In most colonial corals (Van Moorsel 1985) growth does indeed appear to be indeterminate, and there is no evidence of physiological senescence. This is also true of the corals studied here; no decrease in radial growth rate was detected in older corals (Table 4), however a combination of the colony morphology and habitat of the species appear to restrict their potential growth. This observation has major demographic implications, es-

pecially for reproductive schedules, and could not have been readily made if size- or age-dependent measures had been assessed separately.

Adult mortality in all three species was inversely proportional to both colony size (Table 9) and colony age (Fig. 6). Similar trends have been found in all previous studies of coral mortality (Connell 1973, Loya 1976, Rylaarsdam 1983, Bak and Engel 1979, Harriott 1985, Hughes 1984, 1985, Hughes and Jackson 1985). One plausible explanation for this, put forward by Jackson (1979), suggests that the likelihood of any event completely killing a small colony is greater than that for larger colonies, since there is a higher probability of part of a large colony remaining unaffected. As a corollary, it should be expected that although large colonies have a lower rate of total mortality, they will have a higher incidence of partial mortality. This was demonstrated by Hughes and Jackson (1980, 1985) for

TABLE 12. Life table for the coral *Goniastrea favulus*. Size of the 0 age class is based on the estimated fecundity of the populations.  $f_x$  = frequency of genets in each age class; — = below size of visible recruitment; \* = values estimated from adjacent age classes. See Figs. 8 and 9 for standard error of  $l_x$  and  $m_x$ .

| $x$<br>(age class)                                                                                              | $f_x$              | $l_x$                 | $m_x$    | $l_x m_x$<br>( $\bar{X} \pm 1$ SE) |
|-----------------------------------------------------------------------------------------------------------------|--------------------|-----------------------|----------|------------------------------------|
| 0                                                                                                               | $1.91 \times 10^6$ | 1.00                  | 0        | 0                                  |
| 2                                                                                                               | —                  | $2.20 \times 10^{-5}$ | 0        | 0                                  |
| 5                                                                                                               | 127                | $1.65 \times 10^{-5}$ | 930      | .02 $\pm$ .0018                    |
| 10                                                                                                              | 94                 | $1.42 \times 10^{-5}$ | 4500     | .07 $\pm$ .0056                    |
| 15                                                                                                              | 55                 | $1.36 \times 10^{-5}$ | 20 400   | .29 $\pm$ .021                     |
| 20                                                                                                              | 8                  | $1.36 \times 10^{-5}$ | 7810     | .11 $\pm$ .018                     |
| 25                                                                                                              | 0                  | $1.36 \times 10^{-5}$ | 21 600   | .29 $\pm$ .061                     |
| 30                                                                                                              | 2                  | $1.36 \times 10^{-5}$ | 35 300*  | .48 $\pm$ .12                      |
| 35                                                                                                              | 1                  | $1.36 \times 10^{-5}$ | 232 000  | 3.16 $\pm$ .62                     |
| 40                                                                                                              | 0                  | $1.36 \times 10^{-5}$ | 128 000* | 1.75 $\pm$ .32                     |
| 45                                                                                                              | 0                  | $1.36 \times 10^{-5}$ | 128 000* | 1.75 $\pm$ .32                     |
| 50                                                                                                              | 1                  | $1.36 \times 10^{-5}$ | 24 400   | .33 $\pm$ .12                      |
| $\Sigma l_x m_x = 8.24 \pm 0.79$                                                                                |                    |                       |          |                                    |
| $r = 0.061$                                                                                                     |                    |                       |          |                                    |
| generation time ( $G$ ) = 37 + 11 yr, where                                                                     |                    |                       |          |                                    |
| $G = \Sigma x l_x m_x / \Sigma l_x m_x$ ; $r$ = intrinsic rate of increase estimated by iteration (Krebs 1972). |                    |                       |          |                                    |

TABLE 13. Life table for the coral *Platygyra sinensis*. Size of the 0 age class is based on the estimated fecundity of the populations.  $f_x$  = frequency of genets in each age class; - = below size of visible recruitment; \* = values estimated from adjacent age classes. See Figs. 8 and 9 for standard error of  $l_x$  and  $m_x$ .

| $x$<br>(age class)                                                                                              | $f_x$           | $l_x$                 | $m_x$    | $l_x m_x$<br>( $\bar{X} \pm 1 \text{ SE}$ ) |
|-----------------------------------------------------------------------------------------------------------------|-----------------|-----------------------|----------|---------------------------------------------|
| 0                                                                                                               | $1 \times 10^7$ | 1.00                  | 0        |                                             |
| 2                                                                                                               | +               | $1.87 \times 10^{-6}$ | 0        | 0                                           |
| 5                                                                                                               | 12              | $1.21 \times 10^{-6}$ | 0        | 0                                           |
| 10                                                                                                              | 14              | $1.17 \times 10^{-6}$ | 3820     | .0045 $\pm$ .002                            |
| 15                                                                                                              | 16              | $1.11 \times 10^{-6}$ | 97 400   | .114 $\pm$ .05                              |
| 20                                                                                                              | 21              | $0.87 \times 10^{-6}$ | 147 000  | .163 $\pm$ .10                              |
| 25                                                                                                              | 7               | $0.81 \times 10^{-6}$ | 225 000  | .197 $\pm$ .11                              |
| 30                                                                                                              | 8               | $0.81 \times 10^{-6}$ | 337 000  | .274 $\pm$ .14                              |
| 35                                                                                                              | 5               | $0.71 \times 10^{-6}$ | 244 000  | .198 $\pm$ .12                              |
| 40                                                                                                              | 3               | $0.71 \times 10^{-6}$ | 235 000  | .168 $\pm$ .09                              |
| 45                                                                                                              | 1               | $0.71 \times 10^{-6}$ | 108 000  | .077 $\pm$ .04                              |
| 50                                                                                                              | 0               | $0.71 \times 10^{-6}$ | 136 000* | .097 $\pm$ .05                              |
| 55                                                                                                              | 1               | $0.71 \times 10^{-6}$ | 99 300   | .071 $\pm$ .04                              |
| >55                                                                                                             | 1               | $0.71 \times 10^{-6}$ | 103 000  | .074 $\pm$ .04                              |
| $\Sigma l_x m_x = 1.44 \pm 0.025$                                                                               |                 |                       |          |                                             |
| $r = 0.0108$                                                                                                    |                 |                       |          |                                             |
| generation time ( $G$ ) = 34 + 10 yr, where                                                                     |                 |                       |          |                                             |
| $G = \Sigma x l_x m_x / \Sigma l_x m_x$ ; $r$ = intrinsic rate of increase estimated by iteration (Krebs 1972). |                 |                       |          |                                             |

foliose species in Jamaica, and in this study for three massive species (Tables 6 and 10). The rigors of the intertidal habitat may also be responsible for the high rates of colony fission and tissue turnover found in these corals (Tables 6 and 7). Values for tissue turnover are within the range of those for foliose species described by Hughes and Jackson (1985), in non-storm-affected populations. Foliose corals might have been expected to have a much higher turnover than the massive species studied here, due to their more fragile laminar morphology and a relatively poor ability to shed sediments of larger size classes (e.g., *Agaricia*; Hubbard and Pocock 1972).

While the reef-flat habitat may lead to high rates of tissue turnover, *G. aspera*, *G. favulus*, and *P. sinensis* nevertheless appear to display lower rates of colony fission than corals with other morphologies. When the mean annual probabilities of fission (Table 6) were averaged for the six populations of massive corals in this study, a value of 0.029 was obtained, compared with a mean value of 0.05 for eight populations of foliose Caribbean species (Hughes and Jackson 1985; Table 4). It is likely that the rates for open branching species are even greater (Hughes and Connell 1987). These different rates of fission are reflected in the proportion of clonal input to the populations. Levels of clonal input are significant for the populations in this study, with ratios of ramets to genets ranging from 1:1 to over 2:1 (Table 5), however, they are probably lower than those found in corals with other growth forms (Neigel and Avise 1983, Hughes and Jackson 1985, Willis and Ayre 1985).

Asexual or vegetative propagation is an ubiquitous phenomenon in modular organisms (Harper and Bell 1979) and one which enables them to compensate to some extent for the disadvantages of their sessile habit

(Jackson and Hughes 1985). Colony fission has been described as being advantageous in many species of corals since it provides a means for the rapid occupation of space, and spreads the risk of genet mortality (Highsmith 1982, Hughes and Jackson 1985). Highsmith concluded that a number of corals appear to have evolved fragmentation as a part of their life histories, however, the massive faviids on the reef flats of Geoffrey and Pioneer Bays are not among them. Tissue fission in *G. aspera*, *G. favulus*, and *P. sinensis* is probably best considered a form of partial mortality in these slow-growing species, and a cost rather than a benefit. Previously damaged corals were more likely to suffer further damage (Table 8), and age distributions indicated that many older genets were reduced significantly in size (Fig. 5), with some of their component ramets falling below the reproductive threshold. Thus although colonies formed by tissue fission may survive for some time, their combined area perhaps eventually exceeding their original size, fission in these species appeared to entail real costs in terms of fitness, due to increased probability of further damage, as well as a decrease in fecundity related to a decrease in size (Fig. 9).

When compared to the number of new colonies formed by sexual recruitment colony fission made a relatively minor annual contribution to populations in this study, (Tables 1 and 6). The ratio of sexual recruitment to recruitment by fission was higher in all of these massive species (between 14:1 and 3:1) than in any of the foliaceous species studied by Hughes (1988) where ratios ranged from 2:1 down to 1:10. This is true even though several of the foliose species studied by Hughes had quite high levels of sexual recruitment. For example, *Agaricia agaricites* populations had more than twice as many sexual recruits per unit area as did

the highest recruiting population in this study, *G. favulus* at Pioneer Bay, yet *A. agaricites*' ratio of sexual to asexual recruitment was still lower. These differences are likely to be due to the comparatively lower rate of colony fission in massive reef-flat species. These results for massive species extend the trend of relatively low levels of successful sexual reproduction found in other frequently fragmenting or clonally propagating colonial marine taxa and terrestrial plants (Jackson 1985, Wallace 1985, Silvertown 1987).

#### Life tables

**Survivorship.**—Survivorship to visible recruitment was between  $2.6 \times 10^{-6}$  and  $5.3 \times 10^{-5}$  in the three species, a similar rate to that estimated for the gorgonians *Muricea californica* ( $3.5 \times 10^{-7}$ ) and *M. fruticosa* ( $4.9 \times 10^{-6}$ ) on the California coast (Grigg 1977). Estimated overall survivorship was higher for *G. favulus* than for *G. aspera* or *P. sinensis* (Fig. 8), with the major difference between the species occurring in the first age class. The variation is most likely to be the result of dissimilarities during the early life history stages, since mortality rates of newly settled juveniles of all three species in the 1st yr of life were very similar (Babcock 1986). The ability of *G. favulus* to self- as well as cross-fertilize (Heyward and Babcock 1986), and the benthic development of eggs of *G. favulus* may have the effect of reducing egg and larval mortality in the first 1–2 d. A certain degree of self-seeding of *G. favulus* populations may be implied in this conclusion, however biochemical-genetic studies of this species indicate that there is still a high rate of larval exchange between populations (Stoddart et al. 1988). Other recent studies on the dispersal of coral larvae also indicate that transport between reefs occurs at significant levels (Willis and Oliver 1988) and suggest that the majority of settling larvae may come from distant reefs (Babcock 1988). The correlation between adult densities and numbers of recruits found here supports neither argument since it could be the result of a number of factors besides limited dispersal, such as prerecruitment mortality or larval settlement behavior. Provided these assumptions are correct, differences in reproductive behavior appear to have measurable effects on the survivorship of young in these three species.

**Fecundity schedules.**—Components of both age and size determine fecundity, although after adult levels of polyp fecundity are reached, size is of primary importance. Fecundities rose rapidly to quite high levels, but reproductive output decreased in some of the older age classes as a result of decreases in mean surface area of corals, e.g., microatolls or genets that were composed of several ramets (Fig. 9). While age has been shown to affect the level of polyp fecundity in older colonies, which have shrunk to a small size, (Kojis and Quinn 1985, Szmant-Froelich 1985), such effects have not been considered here.

One important consequence of the close linkage of

colony size and colony fecundity in these populations is that of lowering the mean generation times, since the oldest age classes do not make the overwhelming reproductive combination that might be expected if they were also always the largest (e.g., *Muricea californica*; Grigg 1977). It is important, however, to note the variability around these estimates, much of which was the result of size variation among genets in the same age class. For example, the peak in fecundity for *G. favulus* in the 35-yr age class was due to the presence of one large colony, which contributed 12% of the populations' estimated annual reproductive output. This potential dominance of a population's reproductive output by a few genets is similar to that reported in other modular organisms such as trees (Bullock 1982). Given the relatively low level of fission in these massive corals, the potential for reproductive domination of a population by a few genotypes is likely to be lower than in corals with a greater propensity for clonal propagation (e.g., *Porites*; Potts et al. 1985). Therefore, despite the possibility that the generation times obtained here (33–37 yr, Tables 11–13) may be relatively short for corals, the values are equivalent to some of the longest reported, such as those of temperate trees such as *Sequoia* (Bonner 1965). This supports the hypothesis of Potts (1983) that long generation times of corals may be one cause of the low rate of speciation in reef-building corals relative to other marine invertebrate groups in the Indo-Pacific. Long-lived trees such as *Sequoia* are also known to have low rates of speciation (Stebbins 1974).

#### Life history patterns

A comparison of the life history patterns of *G. aspera*, *G. favulus*, and *P. sinensis* suggests that a number of trade-offs are being made in optimizing reproductive success (Table 14). *Goniastrea aspera* was the most fecund species, and had the smallest eggs, while *G. favulus* had the largest eggs, but produced them in low numbers. *Platygyra sinensis* was intermediate in egg size, and also produced eggs in low numbers. The relatively low number of eggs produced by *G. favulus* may be compensated for by rates of prerecruitment mortality in this species, which are lower than in *G. aspera* and *P. sinensis* (Fig. 8). An alternative or additional factor in this difference is that fertilization rates in *G. favulus* may be higher than those of the other two species, due to the ability of *G. favulus* to self-fertilize (Heyward and Babcock 1986). While rates of prerecruitment mortality in both *G. aspera* and *P. sinensis* were similar, *Platygyra* colonies were usually larger and older than those of *G. aspera*. This may compensate to some extent for relatively low egg production per unit area and high larval mortality since *Platygyra* colonies will produce more eggs over a longer period of time. Mean age of populations and age at first reproduction of *P. sinensis* were also greater than for either of the two *Goniastrea* species. *Goniastrea favulus*, with

TABLE 14. Life history traits of *Goniastrea aspera*, *G. favulus*, and *Platygyra sinensis*. Values are means and standard error for combined populations over the entire study.

|                                     | <i>G. aspera</i>                          | <i>G. favulus</i>                          | <i>P. sinensis</i>                        |
|-------------------------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------|
| Egg size ( $\mu$ )                  | 360 $\pm$ 5                               | 420 $\pm$ 7                                | 405 $\pm$ 4                               |
| Fecundity (eggs/cm <sup>2</sup> )   | 940 $\pm$ 260                             | 360 $\pm$ 140                              | 350 $\pm$ 100                             |
| Estimated survivorship to 2 yr      | (1.9 $\pm$ 0.2) $\times$ 10 <sup>-6</sup> | (2.2 $\pm$ 0.04) $\times$ 10 <sup>-5</sup> | (1.9 $\pm$ 0.7) $\times$ 10 <sup>-6</sup> |
| Age at first reproduction (yr)      | 4-5                                       | 5-6                                        | 5-8                                       |
| Median age of adult population (yr) | 9.8                                       | 5.9                                        | 16                                        |
| Estimated mean generation time (yr) | 33 $\pm$ 11                               | 37 $\pm$ 11                                | 34 $\pm$ 10                               |

the lowest prerecruitment mortality, also had the lowest median age of the three species, with *G. aspera* occupying the intermediate position. The fact that these trade-offs do indeed balance out tends to be supported by the similar values for mean generation times in the three species.

The common assumption of a positive correlation between high larval dispersal and high larval mortality in marine invertebrates (Strathmann 1985) appears to be supported by these data. Life history patterns with high levels of juvenile mortality are often associated with long-lived organisms and delayed onset of reproduction (Bell 1976, Stearns 1976, Harper 1977). Trends found within this group of massive corals support this generalization, as does a comparison of life history patterns in a wide range of other scleractinians. *Goniastrea aspera*, *G. favulus*, and *P. sinensis* are all long-lived organisms, which produce large numbers of small eggs. All three species suffer high larval mortality, begin reproduction at  $\approx$  5 yr old, and do not reach adult levels of polyp fecundity (per square centimetre) until they are 10-15 yr old (Fig. 7). Two long-lived species of gorgonian, *Muricea californica* and *M. fruticosa* (Grigg 1977), exhibit patterns of reproduction and mortality very similar to those of *Goniastrea* and *Platygyra*.

In contrast, the solitary coral *Balanophyllia elegans* produces large benthic planulae that settle very close (<1 m) to the parent (Gerrodette 1981). Survivorship for this species is relatively constant with age, reproduction commences at an early age (1.5 yr), and the coral has a life-span of under 10 yr (Fadlallah 1983). Although no estimates of larval mortality are available for *Favia fragum*, this small massive species releases well-developed planulae, many of which are capable of rapid settlement (Lewis 1974). Such larvae are likely to suffer a low mortality in comparison to larvae of other members of the Faviidae such as *Goniastrea* and *Platygyra*, which are gamete spawners. *Favia fragum* reaches reproductive size after 2 yr, and colonies never attain large sizes (Szmant-Froelich et al. 1985). *Stylophora pistillata*, which has been described as a fugitive species (Loya 1976), is another short-lived species that broods planulae. Massive *Porites* species are perhaps the longest lived of all scleractinians (Potts et al. 1985), yet their eggs are among the smallest of all scleractinians (Kojis and Quinn 1982).

Apparently, short-lived species survive by producing rapidly settling larvae with a high probability of sur-

vival. Longer lived species display a contrasting pattern of larval development with longer planktonic dispersal periods. Although risks are inherent in greater dispersal such long-lived species may be able to benefit from its advantages (Palmer and Strathmann 1981). The general trend towards early reproduction and brooding in small, short-lived corals, and for broadcast spawning in larger, longer lived species (Fadlallah 1983, Szmant 1986), is supported by the data presented here. Similar trends have been reported in groups of solitary marine organisms, such as molluscs (Strathmann 1985). Unlike solitary organisms, however, the size of the functional unit or module in colonial organisms such as corals is unrelated to colony size. The fact that this trend is present in modular animals such as corals supports an ecological, rather than a physiological explanation for the correlation between brooding and small adult body size.

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